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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EC503/20EC602/17EC501 – DIGITAL COMMUNICATION

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) A signal $g(t)$ which is bandlimited to W Hz is sampled at a rate f_s . Derive the expression for the sampled version of the signal and also recover the original signal from its samples.
- b) Figure. Q.1.b shows the spectrum of a message signal $g(t)$. The signal is sampled at a rate equal to 1 kHz using flat-top pulses, with each pulse being of 1 volt amplitude and of duration 0.1 ms. Determine and sketch the spectrum of the resulting PAM signal.

10 L*2 1 1

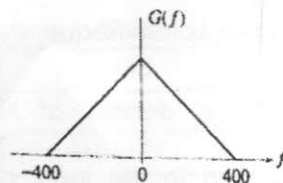


Figure. Q.1.(b)

10 L3 1 1

2. a) A signal $g(t) = 10\cos(20\pi t)\cos(200\pi t)$ is sampled at the rate of 250 samples per second. i) sketch the spectrum of the sampled signal, ii) Specify the cutoff ideal reconstruction filter so as to recover the original signal from the sampled signal. iii) Specify the Nyquist rate for the signal $g(t)$.
- b) Three signals $s_1(t)$, $s_2(t)$, and $s_3(t)$ are as shown in Figure.Q.2.a. Apply Gram-Schmidt procedure to obtain an orthonormal basis for the signals. Express signals $s_1(t)$, $s_2(t)$, and $s_3(t)$ in terms of orthonormal basis functions. Also, give the signal constellation diagram.

10 L3 1 1

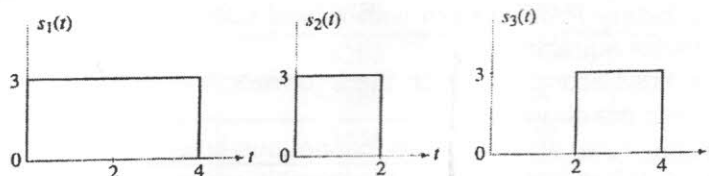


Figure.Q.2.(b)

10 L3 2 1

3. a) A finite energy signal $s(t)$ is shown in Figure.(a) i) Sketch the impulse response $h_{opt}(t)$ of optimum filter matched to signal $s(t)$, ii) Determine the value of the output signal at $t=T$ assuming noise is zero and input is $s(t)$.

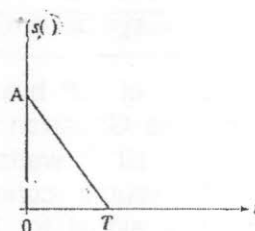


Figure.Q.3(a)

6 L3 2 1

- b) Draw diagrams for the detector and vector receiver parts of the correlation receiver.
- c) With suitable mathematical analysis, show that matched filter receiver is an optimum receiver.

4 L1 2 1

10 L2 2 1

Unit – II

4. a) Derive an expression for the average probability of symbol error in a PCM system.
- b) Derive an expression for the signal-to-quantization noise ratio for midriser quantizer.
- c) For a binary PCM signal, determine L if the compression parameter $\mu = 100$ and the minimum $[SNR]_{0,dB} = 45 \text{ dB}$. Determine the $[SNR]_{0,dB}$ with this value of L.
5. a) For the sinusoidal modulating signal $x(t) = A_0 \cos(2\pi f_0 t)$, show that the output signal-to-quantization noise ratio in a delta modulated system under the assumption of no slope overload is given by

10 L3 2 1

6 L2 3

4 L3 3 1

$$[SNR]_0 = \frac{3f_s^3}{8\pi^2 f_0^2 f_M}$$

where, $f_s = \frac{1}{T_s}$ = sampling frequency and f_M = cutoff frequency of the LPF in the receiver.

6 L2 3 1

- b) Derive an expression for the power spectral density of NRZ Unipolar format.

8 L2 3 1

- c) The binary data stream 011100101 is applied to the input of a correlative coder which has a correlation span of two binary digits.

- i) Construct the coder output and receiver output with a precoder. Assume $b_1=1$, $b_2=1$.

Suppose that owing to error during transmission, the level at the receiver input produced by the third digit is reduced to zero. Construct the new receiver output.

6 L3 4 1

6. a) With a neat diagram of the baseband binary data transmission system, explain intersymbol interference.

8 L2 4 1

- b) A continuous-time signal is converted into a PCM wave. The number of quantization levels = 64. A synchronizing pulse is added at the end of each codeword representing a sample of the analog signal. The resulting PCM wave is sent over a channel of bandwidth 24 kHz using a binary PAM system with raised cosine spectrum having a roll-off factor equal to unity.

- i) Find the rate (in bits/second) at which the information is transmitted through the channel.

- ii) Find the sampling rate at which the continuous-time signal is sampled. What is the maximum possible value for the highest frequency component of the continuous-time signal?

4 2 4 1

- c) With relevant diagrams and mathematical analysis, explain duobinary encoding and decoding.

8 L2 4 1

Unit – III

7. a) Derive an expression for the average probability of symbol error for a coherent binary PSK system.

10 L3 5 1

- b) An FSK system transmits binary data at a rate of 10^6 bits per second. Assuming channel noise is additive white Gaussian with zero mean and power spectral density 2×10^{-20} watts/Hz, determine the average probability of error. Assume coherent detection, and amplitude of received sinusoidal signal for both symbol 1 and 0 to be $1.2 \mu V$.

5 L3 5 1

- c) A binary data is transmitted using ASK over an AWGN channel at a rate of 2.4 MBPS. The carrier amplitude at the receiver is 1 mV and the noise power spectral density is $\frac{N_0}{2} = 10^{-15}$ W/Hz. Find the average probability of error if the detection is coherent. Assume $\text{erfc}(3.5) \cong 3 \times 10^{-6}$.
8. a) Explain M-ary QAM technique with neat diagrams for the constellation diagram with M=16. Also, obtain an expression for the average probability of symbol error.
- b) In a QPSK modulation scheme, the input binary sequence $\{b_i\} = \{11001011101\}$. The bit interval is T_b . Give the demultiplexed sequence $\{b_{ei}\}$ and $\{b_{oi}\}$. Assuming NRZ polar format, sketch the sequences $\{b_i\}$, $\{b_{ei}\}$ and $\{b_{oi}\}$.
- c) With relevant mathematical expressions, construct the constellation diagram for a QPSK system.

5 L3 5 1

10 2 5 1

5 L3 5 1

5 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

Table of Error Function

u	$\text{erf}(u)$	u	$\text{erf}(u)$
0.00	0.00000	1.10	0.88021
0.05	0.05637	1.15	0.89612
0.10	0.11246	1.20	0.91031
0.15	0.16800	1.25	0.92290
0.20	0.22270	1.30	0.93401
0.25	0.27633	1.35	0.94376
0.30	0.32863	1.40	0.95229
0.35	0.37938	1.45	0.95970
0.40	0.42839	1.50	0.96611
0.45	0.47548	1.55	0.97162
0.50	0.52050	1.60	0.97635
0.55	0.56332	1.65	0.98038
0.60	0.60386	1.70	0.98379
0.65	0.64203	1.75	0.98667
0.70	0.67780	1.80	0.98909
0.75	0.71116	1.85	0.99111
0.80	0.74210	1.90	0.99279
0.85	0.77067	1.95	0.99418
0.90	0.79691	2.00	0.99532
0.95	0.82089	2.50	0.99959
1.00	0.84270	3.00	0.99998
1.05	0.86244	3.30	0.99999
